



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2025

MATHEMATICS: PAPER I

MARKING GUIDELINES

Time: 3 hours

150 marks

These marking guidelines are prepared for use by examiners and sub-examiners, all of whom are required to attend a standardisation meeting to ensure that the guidelines are consistently interpreted and applied in the marking of candidates' scripts.

The IEB will not enter into any discussions or correspondence about any marking guidelines. It is acknowledged that there may be different views about some matters of emphasis or detail in the guidelines. It is also recognised that, without the benefit of attendance at a standardisation meeting, there may be different interpretations of the application of the marking guidelines.

SECTION A**QUESTION 1**

(a) Solve for x .

(1) $3x(x-2)(4x+7)=0$

$$x=0 \quad \text{or} \quad x=2 \quad \text{or} \quad x=-\frac{7}{4}$$

(2) Solve for x , correct to **two decimal places** by completing the square.

$$x^2 - 2x - 2 = 0$$

$$x^2 - 2x + (-1)^2 = 2 + (-1)^2$$

$$(x-1)^2 = 3$$

$$x-1 = \pm\sqrt{3}$$

$$x = 1 \pm \sqrt{3}$$

$$x = 2,73 \quad \text{or} \quad x = 0,73$$

(3) Solve for x : $x^2 - 2x - 8 < 0$ if $x \in \mathbb{Z}$.

$$x^2 - 2x - 8 < 0$$

$$(x-4)(x+2) < 0$$

$$-2 < x < 4$$

$$\therefore \{-1; 0; 1; 2; 3\}$$

- (b) Show that the equation $x^2 - 2px = -p^2$ has equal roots for all values of p .

$$\begin{aligned}
 x^2 - 2px &= -p^2 \\
 x^2 - 2px + p^2 &= 0 \\
 \Delta &= b^2 - 4ac \\
 &= (-2p)^2 - 4(1)(p^2) \\
 &= 4p^2 - 4p^2 \\
 &= 0
 \end{aligned}$$

ALT

$$\begin{aligned}
 x^2 - 2px + p^2 &= 0 \\
 (x - p)(x - p) &= 0 \\
 x &= p
 \end{aligned}$$

QUESTION 2

- (a) Solve for x , if $2^x + 2^{x+1} = 192$.

$$\begin{aligned}
 2^x + 2^{x+1} &= 192 \\
 2^x(1 + 2) &= 192 \\
 2^x &= 64 \\
 2^x &= 2^6 \\
 x &= 6
 \end{aligned}$$

- (b) Solve for x if $\sqrt{5-x} + x + 1 = 0$.

$$\begin{aligned}
 \sqrt{5-x} &= -x - 1 \\
 5 - x &= x^2 + 2x + 1 \\
 0 &= x^2 + 3x - 4 \\
 0 &= (x + 4)(x - 1)
 \end{aligned}$$

$$x = -4 \quad \text{or} \quad x = 1$$

- (c) If $\log 2 = p$ and $\log 3 = q$ determine the value of $\log 1\frac{1}{2}$ in terms of p and q .

$$\begin{aligned} &= \log \frac{3}{2} \\ &= \log 3 - \log 2 \\ &= q - p \end{aligned}$$

- (d) If $\log x + \log z = \log y^2$, show that x , y and z form a geometric sequence.

$$\begin{aligned} \log x + \log z &= \log y^2, \\ \log xz &= \log y^2 \checkmark \\ xz &= y^2 \checkmark \end{aligned}$$

QUESTION 3

- (a) Given that: $f(x) = -2x^2 - 2$.

- (1) Determine $\frac{f(x+h) - f(x)}{h}$, the average rate of change of f over the interval $[x; x+h]$. **Simplify your answer.**

$$\begin{aligned} m_{\text{ave}} &= \frac{-2(x+h)^2 - 2 - (-2x^2 - 2)}{h} \\ &= \frac{-2(x^2 + 2xh + h^2) - 2 + 2x^2 + 2}{h} \\ &= \frac{-2x^2 - 4xh - 2h^2 - 2 + 2x^2 + 2}{h} \\ &= \frac{-4xh - 2h^2}{h} \\ &= \frac{h(-4x - 2h)}{h} \\ &= -4x - 2h \end{aligned}$$

- (2) Hence, determine $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$\begin{aligned} &= \lim_{h \rightarrow 0} -4x - 2h \\ &= -4x \end{aligned}$$

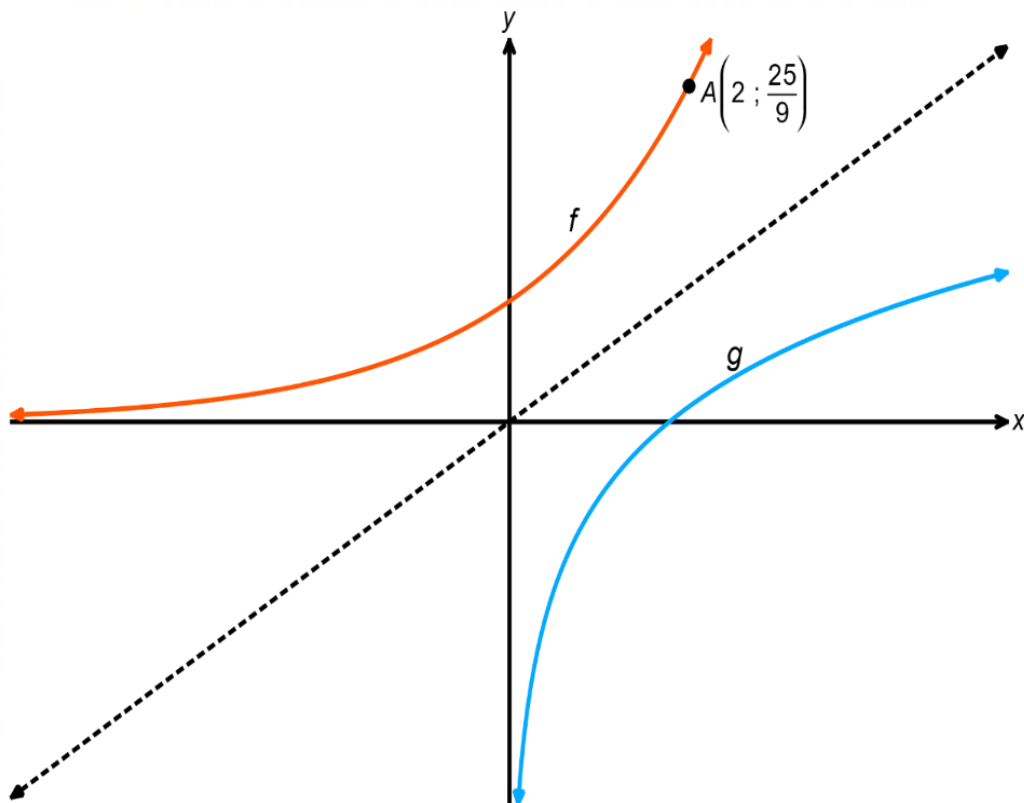
(b) Find $f'(x)$, if $f(x) = \sqrt[3]{x^2} - \frac{3}{x^5} + \pi^2$

$$f(x) = \underline{x^{\frac{2}{3}} - 3x^{-5}} + \pi^2$$

$$f'(x) = \frac{2}{3}x^{-\frac{1}{3}} + 15x^{-6} + 0$$

(c) In the diagram below:

- $f(x) = p^x$.
- $A\left(2; \frac{25}{9}\right)$ lies on the graph of f .
- f and g are symmetrical about the line $y = x$.



- (1) Determine the value of p .

$$f(x) = p^x$$

$$\frac{25}{9} = p^2$$

$$p = \frac{5}{3}$$

- (2) Determine the equation of g .

$$g(x) = \log_{\frac{5}{3}} x$$

- (3) Write down the domain and range of f .

$$\text{Domain: } x \in \mathbb{R}$$

$$\text{Range: } y > 0$$

- (4) (i) h is formed if f is moved two units to the right. Write down the equation of $h(x)$.

$$h(x) = \left(\frac{5}{3}\right)^{x-2}$$

- (ii) What is the equation of the asymptote of h ?

$$y = 0$$

QUESTION 4

- (a) Given that: $S_n = \frac{n}{2}[2a + (n-1)d]$.

Determine a in terms of S_n , n and d .

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$2S_n = n[2a + (n-1)d]$$

$$2S_n = 2an + n(n-1)d$$

$$2S_n - n(n-1)d = 2an$$

$$a = \frac{2S_n - n(n-1)d}{2n}$$

- (b) In an arithmetic series, $S_{10} = 500$ and the common difference is $\frac{100}{9}$.
Determine the first term.

$$a = \frac{2S_n - (n-1)d}{2n}$$

$$a = \frac{2(500) - 10(10-1)\left(\frac{100}{9}\right)}{2(10)}$$

$$a = 0$$

ALT

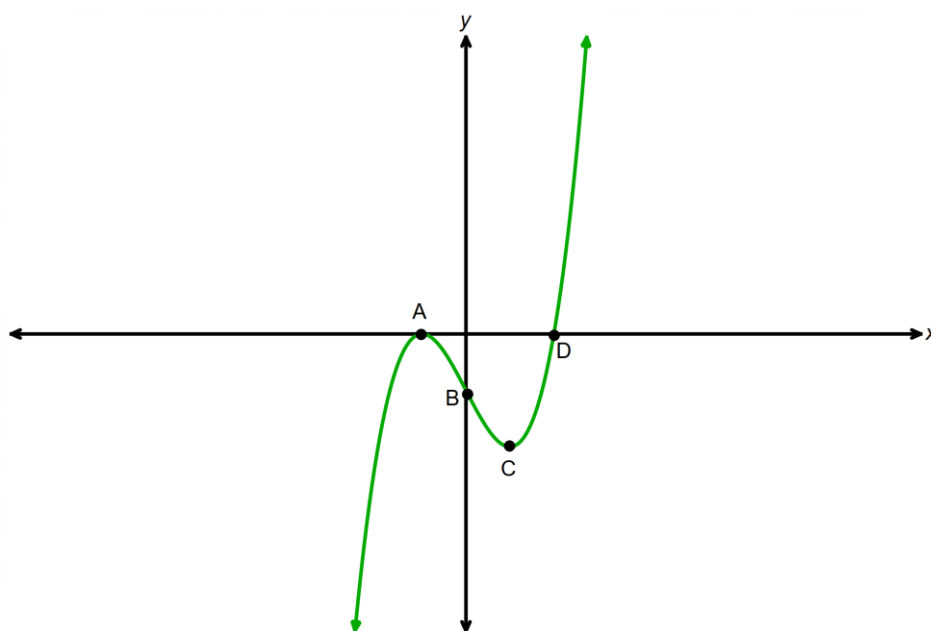
$$500 = \frac{10}{2} \left[2a + (10-1)\frac{100}{9} \right]$$

$$a = 0$$

QUESTION 5

In the diagram below:

- $g(x) = x^3 - 3x - 2$.
- A and C are the turning points of g .
- A and D are the x -intercepts.
- B is the y -intercept.



- (a) Determine the coordinates of B and C.

$$g(0) = -2$$

$$B(0; -2)$$

$$g'(x) = 3x^2 - 3$$

$$0 = 3(x^2 - 1)$$

$$0 = (x - 1)(x + 1)$$

$$x = 1 \text{ or } x = -1$$

$$g(1) = (1)^3 - 3(1) - 2$$

$$g(1) = -4$$

$$C(1; -4)$$

- (b) Find the x coordinate(s) of the point(s) where $2y - 18x = k$ is a tangent to g .

$$2y - 18x = k$$

$$2y = 18x + k$$

$$y = 9x + \frac{k}{2}$$

$$3x^2 - 3 = 9$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

QUESTION 6

Give the correct answer rounded off to two decimal places.

- (a) Tlhogi establishes a trucking company. The first truck he buys costs R750 000, and it depreciates at 25% per annum using a reducing balance method. However, due to inflation, the price of a new truck appreciates at a rate of 10% per annum.

- (1) Calculate the depreciated value of the truck after 4 years.

$$A = P(1 - i)^n$$

$$= 750000(1 - 0,25)^4$$

$$= 237304,69$$

- (2) Calculate the value of a new truck in 4 years' time.

$$\begin{aligned} A &= P(1+i)^n \\ &= 750000(1+0,1)^4 \\ &= 1098075 \end{aligned}$$

- (3) Tlhogi plans to replace the original truck in 4 years, using it as a trade-in and to pay the remaining amount in cash. To save up for the required cash amount he makes equal monthly payments into a sinking fund earning interest of 8,5% per annum, compounded monthly. He makes his first payment at the end of the first month of the 4 years and his last at the end of the 4 years.

Calculate the value of each monthly instalment he pays into the sinking fund.

$$\begin{aligned} \text{cash} &= (2) - (1) \\ &= 860770,31 \end{aligned}$$

$$\begin{aligned} 860770,31 &= \frac{x \left[\left(1 + \frac{8,5}{1200} \right)^{48} - 1 \right]}{\frac{8,5}{1200}} \\ &= 15119,40 \end{aligned}$$

- (b) An amount invested at x % per annum, compounded monthly, doubles in 10 years. Determine the value of x .

$$\begin{aligned} 2P &= P \left(1 + \frac{x}{1200} \right)^{10 \times 12} \\ 2 &= \left(1 + \frac{x}{1200} \right)^{120} \\ 1 + \frac{x}{1200} &= \sqrt[120]{2} \\ x &= 6,95\% \end{aligned}$$

SECTION B**QUESTION 7**

- (a) The first four terms of a quadratic number pattern are 4, 7, $x+7$ and $x+y+7$. If the fourth term is 58, determine the values of x and y .

$$x + y + 7 = 58$$

$$x + y = 51$$

$$\begin{array}{cccc} 4 & ; & 7 & ; & x+7 & ; & 58 \\ & & 3 & & x & & 51-x \\ & & x-3 & & & & 51-2x \end{array}$$

$$x - 3 = 51 - 2x$$

$$\therefore x = 18$$

$$y = 33$$

- (b) Determine $\sum_{k=0}^{50} \left((-3)^k + 3k \right)$.

$$\sum_{k=0}^{50} (-3)^k = 1 - 3 + 9 + \dots + (-3)^{50}$$

$$\begin{aligned} S_{51} &= \frac{1 - (-3)^{51}}{1 + 3} \\ &= \frac{1 + 3^{51}}{4} \end{aligned}$$

$$\sum_{k=0}^{50} 3k = 0 + 3 + 6 + \dots + 150$$

$$\begin{aligned} S_{51} &= \frac{51}{2} [2(0) + (51-1)(3)] \\ &= 3825 \end{aligned}$$

$$\sum_{k=0}^{50} \left((-3)^k + 3k \right) = \frac{1 + 3^{51}}{4} + 3825$$

Alternative 1

$$4; \quad 7; \quad x+7; \quad x+y+7$$

$$\frac{3}{x-3} \quad \frac{x}{y-x}$$

$$2a = x - 3 \quad 3a + b = 3 \quad a + b + c = 4$$

$$58 = a(4)^2 + b(4) + c$$

$$58 = 16a + 4b + c$$

$$58 = 16\left(\frac{x-3}{2}\right) + 4\left(-\frac{3}{2}x + \frac{15}{2}\right) + x - 2$$

$$x = 18$$

$$y = 2(18) - 3$$

$$y = 33$$

QUESTION 8

Given that: $f(x) = (x+1) + (x+1)^2 + (x+1)^3 + \dots + (x+1)^k + \dots$

If it is given that f is a converging series.

(a) Simplify f in terms of x .

$$f(x) = \frac{x+1}{1-(x+1)}$$

$$= \frac{x+1}{1-x-1}$$

$$= \frac{x+1}{-x}$$

(b) Determine the domain of f .

$$-1 < x+1 < 1$$

$$-2 < x < 0$$

(c) Determine $f'\left(-\frac{1}{2}\right)$.

$$f(x) = \frac{x+1}{-x}$$

$$= -1 - \frac{1}{x}$$

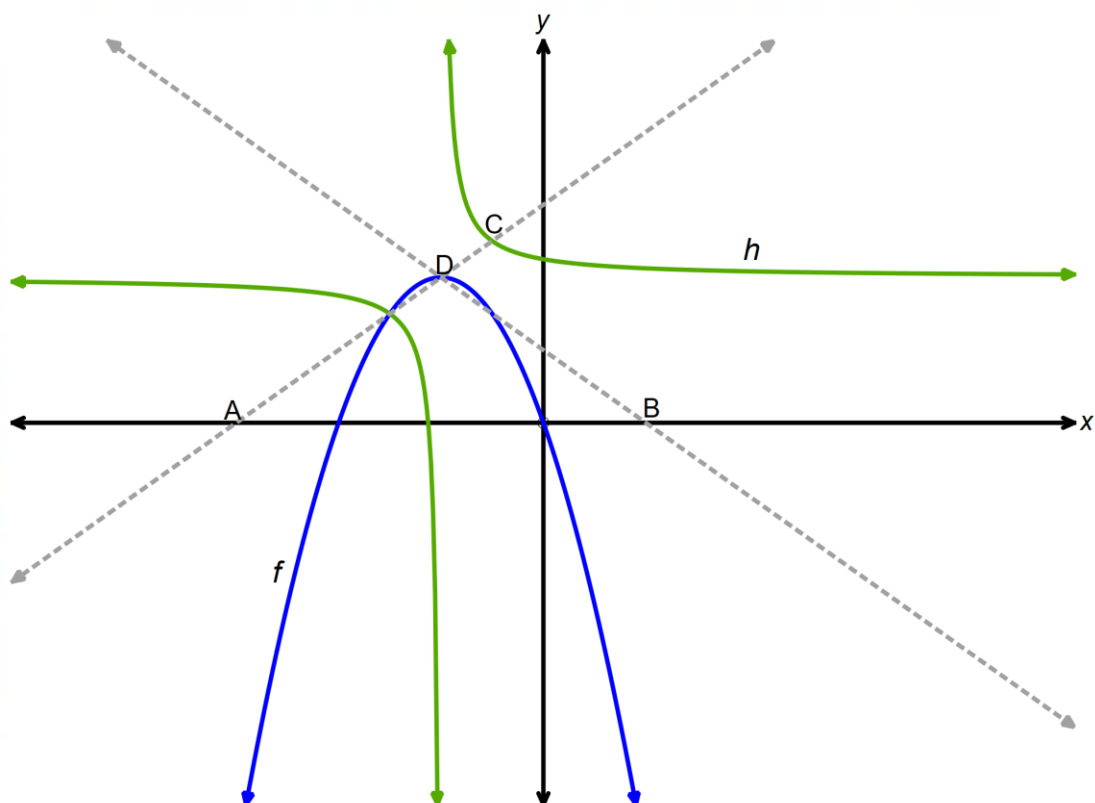
$$f'(x) = \frac{1}{x^2}$$

$$f'\left(-\frac{1}{2}\right) = 4$$

QUESTION 9

In the diagram below:

- $f(x) = a(x - m)^2 + k$ and $h(x) = \frac{1}{x - p} + q$.
- The equations of the axes of symmetry of h are $y = -x + 2$ and $y = x + 6$.
- D is the turning point of f and lies on both equations of the axes of symmetry of h .
- C is a point of intersection of h and the equation of the axis of symmetry of h .
- A and B are x-intercepts.
- The graph of f passes through the origin.



- (a) Determine the domain of h .

$$-x + 2 = x + 6$$

$$-2x = 4$$

$$x = -2$$

$$x \in R : x \neq -2$$

- (b) Determine the values of a , m and k .

$$m = -2$$

$$k = 4$$

$$0 = a(0 + 2)^2 + 4$$

$$0 = 4a + 4$$

$$a = -1$$

- (c) Find the coordinates of A and B.

$$0 = -x + 2$$

$$x = 2$$

$$B(2; 0)$$

$$0 = x + 6$$

$$x = -6$$

$$A(-6; 0)$$

(d) Determine the area of $\triangle ABC$.

$$h(x) = \frac{1}{x+2} + 4$$

$$y = x + 6$$

$$\frac{1}{x+2} + 4 = x + 6$$

$$1 + 4x + 8 = x^2 + 8x + 12$$

$$0 = x^2 + 4x + 3$$

$$0 = (x+3)(x+1)$$

$$x = -3 \quad \text{or} \quad x = -1$$

$$y = -1 + 6$$

$$y = 5$$

$$C(-1; 5)$$

$$\begin{aligned}\triangle ABC &= \frac{1}{2}(8) \times (5) \\ &= 20\end{aligned}$$

QUESTION 10

(a) If $g(x) = -a^{-x} + k$ then:

- (1) Determine the coordinates of the x-intercept of the graph of g in terms of a and k .

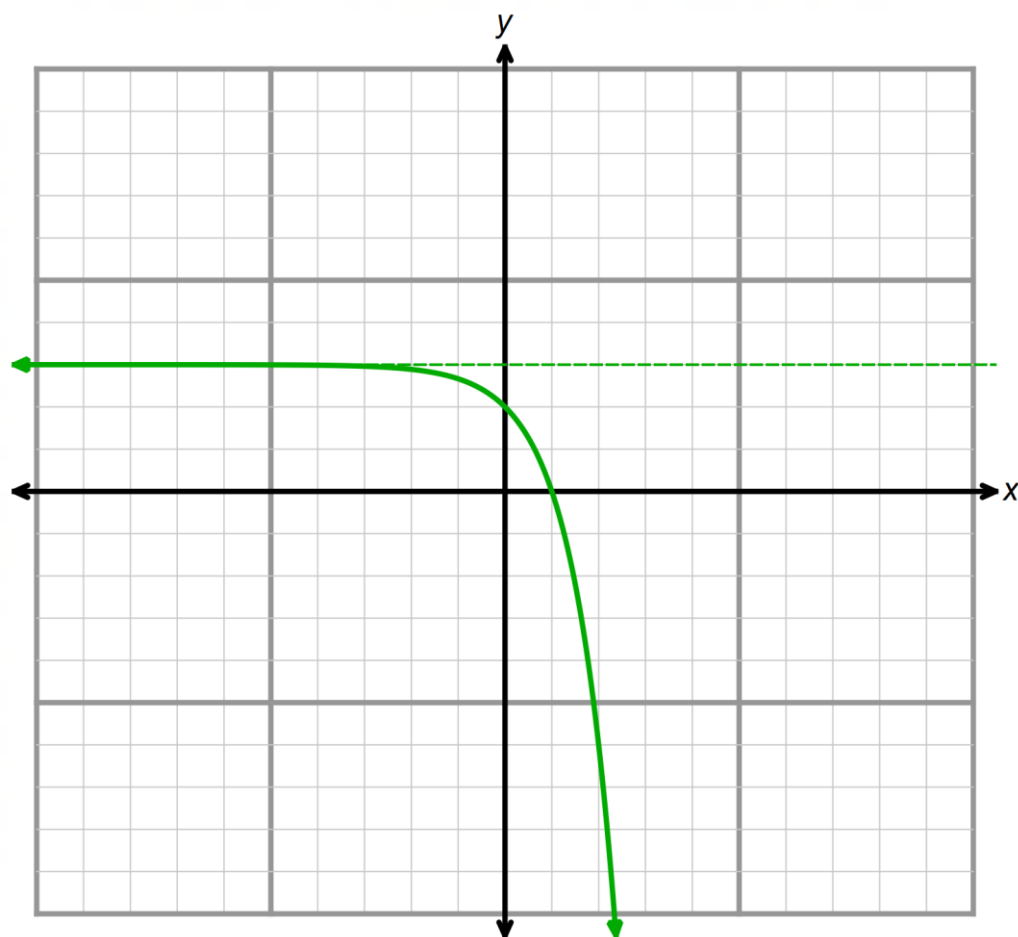
$$0 = -a^{-x} + k$$

$$k = a^{-x}$$

$$-x = \log_a k$$

$$x = -\log_a k$$

- (2) Sketch the graph of $y = g(x)$ if $0 < a < 1$ and $k > 2$.



(b) Given that: $f = \{(-1; 2); (2; -1); (4; 7); (-3; 1); (5; 6)\}$

(1) Is f a function? Explain.

Yes, each input has exactly one output, so this is a function.

(2) Give the domain of f^{-1} .

$$\{2, -1, 7, 1, 6\}$$

(3) Determine p if $f[f^{-1}(2)] = p$.

$$f^{-1}(2) = -1$$

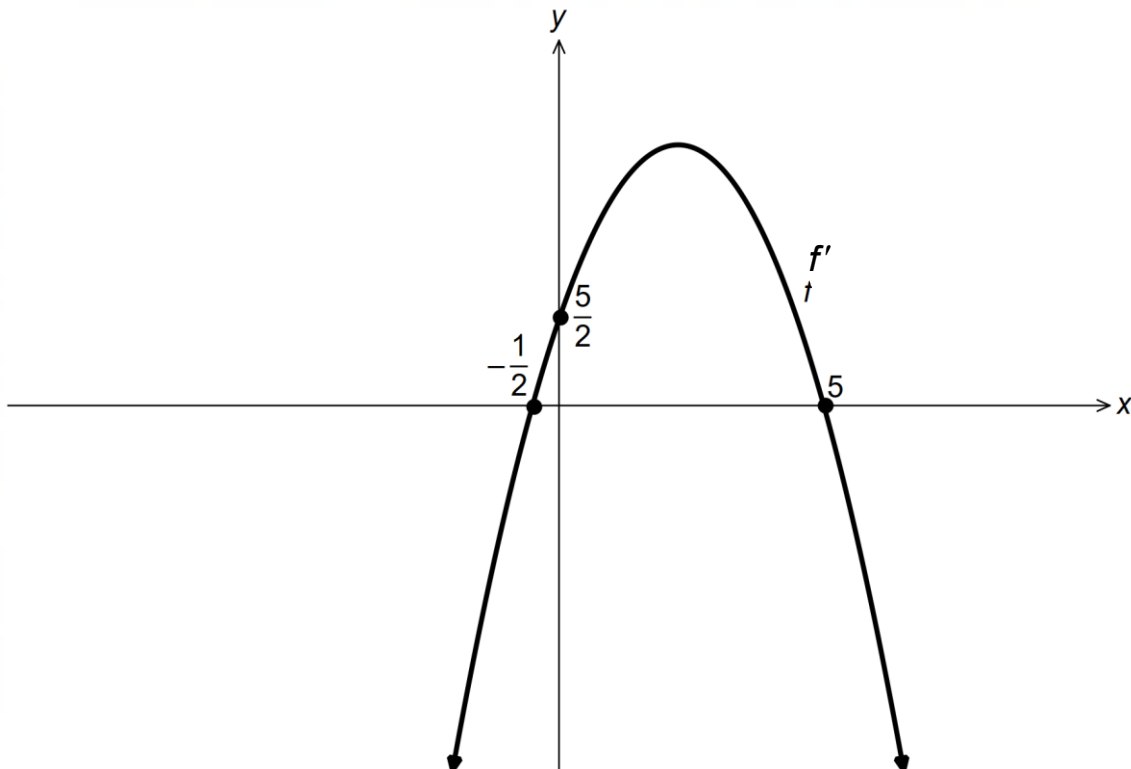
$$f(-1) = 2$$

$$p = 2$$

QUESTION 11

In the diagram below, the graph of $y = f'(x)$ the derivative of $y = f(x)$ is drawn.

- $x = -\frac{1}{2}$ and $x = 5$ are x -intercepts of f' .
- $y = \frac{5}{2}$ is the y -intercept of f' .



- (a) Determine the values of x for which f is increasing.

$$-\frac{1}{2} < x < 5$$

- (b) Determine the value of x for which $f''(x) = 0$.

$$x = \frac{-\frac{1}{2} + 5}{2}$$

$$x = \frac{9}{4}$$

- (c) Determine the equation of the tangent to the curve f at the point $(0; 9)$.

$$y = \frac{5}{2}x + c$$

$$y = \frac{5}{2}x + 9$$

- (d) For which values(s) of x is $\frac{f'(x)}{2x} \leq 0$?

$$-\frac{1}{2} \leq x < 0$$

or

$$x \geq 5$$

QUESTION 12

A local food packaging company in South Africa produces canned food in the shape of cylinders, each with a fixed volume of $150\pi \text{ cm}^3$. To reduce production costs, the company aims to minimise the total surface area of each can.

Given that r represents the radius and h the can's height, determine the radius, which will result in minimum surface area.

Useful formulae

$$S = 2\pi r^2 + 2\pi rh$$

$$V = \pi r^2 h$$

$$\pi r^2 h = 150\pi$$

$$r^2 h = 150$$

$$h = \frac{150}{r^2}$$

$$S = 2\pi r^2 + 2\pi rh$$

$$S = 2\pi r^2 + 2\pi r \left(\frac{150}{r^2} \right)$$

$$= 2\pi r^2 + \frac{300\pi}{r}$$

$$\frac{dS}{dr} = 4\pi r - 300\pi r^{-2}$$

$$4\pi r - 300\pi r^{-2} = 0$$

$$4r^3 = 300$$

$$r = \sqrt[3]{75}$$

Alternative 1

$$\pi r^2 h = 150\pi$$

$$r^2 = \frac{150}{h} = \sqrt{\frac{150}{h}}$$

$$s(h) = 2\pi \left(\sqrt{\frac{150}{h}} \right) + 2\pi \left(\sqrt{\frac{150}{h}} \right) h = 300\pi h^{-1} + 2\pi \sqrt{150} h^{\frac{1}{2}}$$

$$\frac{ds}{dh} = -300\pi h^{-2} + \pi \sqrt{150} h^{-\frac{1}{2}}$$

$$\sqrt{150} h^{\frac{3}{2}} = 300$$

$$h = \sqrt[3]{600}$$

$$r = \sqrt[3]{75}$$

QUESTION 13

(a) In a group of 21 students, the following is true:

- 11 have taken History
- 6 have taken History but not Geography

(1) Determine the number of students who have taken History and Geography.

$$\begin{aligned} H \text{ and } G &= 11 - 6 \\ &= 5 \end{aligned}$$

(2) Determine the maximum number of students who have taken Geography but not History.

The maximum is **10 students**.

(3) If 7 students take Geography, are the events 'Taking History as a Subject' and 'Taking Geography as a Subject' independent or not?

H is the event of taking **History**.

G is the event of taking **Geography**.

$$P(H \text{ and } G) = \frac{5}{21}$$

$$P(H) \times P(G) = \frac{11}{21} \times \frac{7}{21}$$

$$P(H \text{ and } G) \neq P(H) \times P(G)$$

The events 'Taking History' and 'Taking Geography' **are not independent**.

(b) A password uses two letters of the alphabet, followed by four digits from 0 to 9, where repetition of letters and digits is allowed. What is the probability that the code starts with a vowel and ends with an even digit greater than 4?

$$n(s) = 26 \times 26 \times 10 \times 10 \times 10 \times 10$$

$$n(E) = 5 \times 26 \times 10 \times 10 \times 10 \times 2$$

$$\begin{aligned} P(E) &= \frac{5 \times 26 \times 10 \times 10 \times 10 \times 2}{26 \times 26 \times 10 \times 10 \times 10 \times 10} \\ &= \frac{1}{26} \end{aligned}$$

Alternative 1

Probability first letter is a vowel:

$$\frac{5}{26}$$

The probability first letter is 6 or 8:

$$\frac{2}{10}$$

$$\begin{aligned}\therefore P(E) &= \frac{5}{26} \times \frac{1}{5} \\ &= \frac{1}{26}\end{aligned}$$

(c) Given that: A and B are two events in a sample space, S.

- $n(S) = 10$ and $n(A) = 9$.
- $P(B') = 0,2$

Masego argues that the probability of A and B is at least 70%. Do you agree with this claim? **Justify your answer with reasoning and calculations.**

We have to show that: $P(A \text{ and } B) \geq 0,7$

$$\left. \begin{array}{l} P(A) = 0,9 \\ P(B) = 1 - 0,2 \\ \quad = 0,8 \end{array} \right\}$$

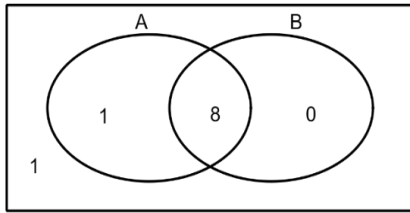
$$\begin{aligned}P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ P(A \text{ and } B) &= P(A) + P(B) - P(A \text{ or } B) \\ P(A \text{ and } B) &= 0,9 + 0,8 - P(A \text{ or } B) \\ P(A \text{ and } B) &= 1,7 - P(A \text{ or } B) \quad \text{but } 0 \leq P(A \text{ or } B) \leq 1\end{aligned}$$

$$P(A \text{ and } B) \geq 1,7 - 1 = 0,7$$

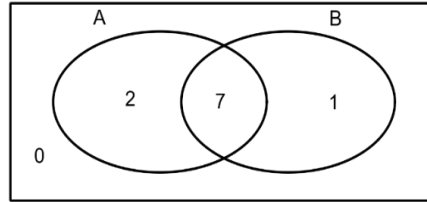
Masego's claim is correct.

Alternative 1

OPTION 1



OPTION 2



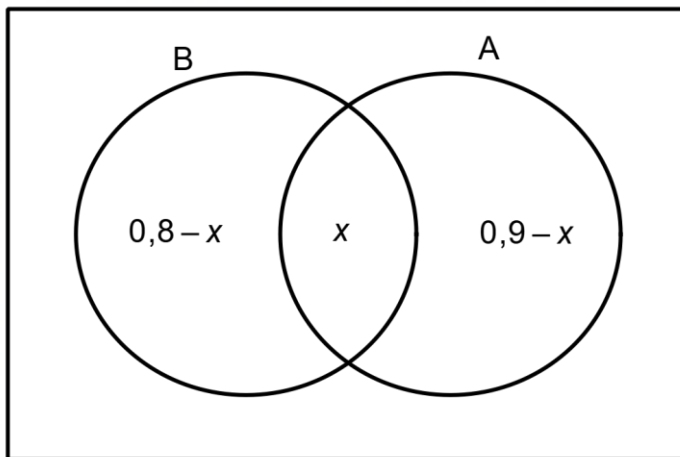
Option 1 Option 2

$$\left. \begin{array}{l} P(A) = 0,9 \\ P(B) = 1 - 0,2 \\ \quad = 0,8 \end{array} \right\}$$

Masego's claim is correct.

Alternative 2

$$\left. \begin{array}{l} P(A) = 0,9 \\ P(B) = 1 - 0,2 \\ \quad = 0,8 \end{array} \right\}$$



Venn Diagram

$$\begin{aligned} 0,8 - x + x + 0,9 - x &\leq 1 \\ -x &\leq -0,7 \\ x &\geq 0,7 \end{aligned}$$

Masego's claim is correct.

Total: 150 marks